MIL-B-60319 (MU)
2 APRIL 1965
SUPERSEDING
(SEE 6.4)

MILITARY SPECIFICATION

BURSTER, PROJECTILE, M19 LOADING, ASSEMBLING AND PACKING

1. SCOPE

1.1 This specification covers the loading, assembling and packing for one type of burster designated as Burster, Projectile, M10.

2. APPLICABLE DOCUMENTS

3.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-2550 — Ammunition and Special Weapons, General Specification for.

MIL-I-45607 — Inspection Equipment, Supply and Maintenance of.

STANDARDS

MILITARY

MIL-STD-105 — Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105.)

MIL-STD-109 — Quality Assurance Terms and Definitions.

MIL-STD-1235 — Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes.

DRAWINGS

ORDNANCE CORPS

9205343 — Burster, Projectile M19 Assembly.

8799212 — Carton, Packing, Ammunition, for Burster, Projectile, M19.

8799213 — Box, Packing, Ammunition, for Burster, Projectile, M19.

8796522 — Marking Diagram and Sealing for Wood Packing Boxes.

FSC 1310

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PUBLICATIONS

ORDNANCE CORPS

IEL–9205343 — Index of Inspection Equipment Lists.

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material. Materials shall be in accordance with applicable drawings and specifications..

3.2 Burster. The burster shall comply with all requirements specified on Drawing (dwg.) 9205343 and with all requirements specified in applicable specifications.

3.3 Hydrostatic test. The burster tubing shall withstand the application of a minimum (min.) hydrostatic pressure without rupture or indication of flaws, as evidenced by leaks, and without permanent deformation beyond drawing requirements, when tested as specified in 4.3.2 (see 6.5).

3.4 Workmanship. All parts shall be free from burrs, chips, cracks, dirt, grease, rust and other foreign material. The cleaning method shall not be injurious to any of the parts nor shall the parts be contaminated by the cleaning agent.

4. QUALITY ASSURANCE PROVISIONS

4.1 General quality assurance provisions. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to MIL–STD–109 in order to define terms used herein. The provisions of MIL–A–2550 shall apply.

4.1.1 *Submission of product.* At the time the completed lot of product is submitted to the Government for acceptance the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

(a) A statement that the lot complies with all of the quality assurance provisions specified within this specification.

(b) Number of units of product inspected.

(c) Results obtained, by defect code, for all inspection performed.

(d) Drawing, specification number and date, together with an identification and date of changes.

(e) Certificates of compliance on all material purchased by the contractor when such material is controlled by Government specifications referenced in any of the contractual documents.

(f) Number of items in the lot.

(g) Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is

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vested in another agent of the certifying organization.

4.1.2 *Government verification.* Using the contractor's written quality assurance procedure, this detail specification, the applicable drawings and other contractual documents as a guide, the Government inspector shall verify at unscheduled intervals all quality assurance operations performed by the contractor. Verification will be performed to the extent necessary to assure compliance with the contractual requirements. Severity of Government inspection of individual characteristics will be directly related to the seriousness of the classification assigned. In no instance will a characteristic classified "critical" be accepted solely on the basis of the contractor's records.

4.2 Inspection provisions.

4.2.1 *Lot formation.* A lot shall consist of charge burster assemblies produced by one manufacturer in one unchanged process, in accordance with the same drawing, same drawing revision, same specification, and same specification revision. Drawing, specification, and process changes not affecting safety, performance or fit as determined by the Government shall not necessitate changing the lot interfix number. Product shall be submitted for inspection in accordance with MIL–STD–105 (or MIL–STD–1235, when applicable). Each lot shall contain:

(a) RDX from not more than one lot from one manufacturer.

(b) Parts of one lot interfix number from one manufacturer.

4.2.2 *Examination.* Inspection for Critical defects shall be 100 percent. Sampling plans and procedures for Major and Minor defects shall be in accordance with MIL–STD–105 except that continuous sampling plans in accordance with Standard MIL–STD–1235 may be used if approved by the procuring activity. Also, at the option of the procuring activity, AQL's and sampling plans may be applied to individual characteristics listed using an AQL of 0.65 percent for each Minor defect and an AQL of 0.40 percent for each Major defect. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.2.4.

4.2.2.1 Tube, Burster, prior to assembly (see dwg. 9205342 covering a detail of dwg. 9205343).

Categories	*Defects*	*Method of Inspection*	*Code No.*
Critical:	None defined.		
Major:	AQL 0.65 percent		
101.	Total length	Gage	01001
102.	Inside diameter	Gage	01002
103.	Wall thickness	Gage	01003
104.	Tube curvature excessive	Gage	01004
Minor:	AQL 1.00 percent		
201.	Finish improper or missing	Visual	01005
202.	Evidence of poor workmanship (see 3.4)	Visual	01006

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4.2.2.2 Assembly (prior to securing discs) (see dwg. 9205343).

Categories	*Defects*	*Method of Inspection*	*Code No.*
Critical:			
1.	Explosive above flush	Gage	02001
2.	Explosive more than .02 inch below end of tube (both ends)	Gage	02002
Major:	None defined.		
Minor:	None defined.		

4.2.2.3 Assembly (see dwg. 9205343).

Categories	*Defects*	*Method of Inspection*	*Code No.*
Critical:	None defined.		
Major:	AQL 1.00 percent		
101.	Total length, maximum (max.)	Gage	03001
102.	Chamber gage failure	Gage	03002
103.	Weight of charge	Balance	03003
104.	Charge loose	Manual	03004
105.	Crack in tube	Visual	03005
106.	Coating damaged, bare area except slight scratches (when applicable)	Visual	03006
Minor:	AQL 1.00 percent		
201.	Closing disc torn or improperly sealed	Visual	03007
202.	Evidence of poor workmanship (see 3.4)	Visual	03008

4.2.2.4 Carton prior to sealing (see dwg. 87 99212).

Categories	*Defects*	*Method of Inspection*	*Code No.*
Critical:	None defined.		
Major:	AQL 0.40 percent		
101.	Burster projectile missing	Visual	04001
Minor:	AQL 0.65 percent		
201.	Any packing component missing	Visual	04002

4.2.2.5 Carton sealed (see dwg. 8799212).

Categories	*Defects*	*Method of Inspection*	*Code No.*
Critical:	None defined.		
Major:	AQL 0.40 percent		
101.	Tape improperly applied	Visual	05001

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Categories	*Defects*	*Method of inspection*	*Code No.*
Minor:	AQL 1.0 percent		
201.	Marking misleading, missing or unidentifiable	Visual	05002
202.	Contents loose upon shaking	Manual	05003

4.2.2.6 Sealed wood packing box (see dwg. 8799213).

Categories	*Defects*	*Method of Inspection*	*Code No.*
Critical:	None defined.		
Major:	AQL 0.40 percent		
101.	Box damaged	Visual	06001
102.	Strapping missing, broken or loose	Visual-Manual	06002
Minor:	AQL 1.50 percent		
201.	Strapping improperly assembled	Visual	06003
202.	Lid improperly assembled	Visual	06004
203.	Markings misleading, missing or unidentifiable	Visual	06005
204.	Contents loose in box	Manual	06006

4.2.3 *Testing.*

4.2.3.1 *Determination of binder lubricant content of RDX (see dwg. 8846601) Major defect, Code No. 02003.* A five gram sample of the explosive shall be selected for test during each eight hours production. The sample shall be taken from the loading line at the time of loading. If the binder lubricant content of the explosive exceeds that specified on the applicable drawing, the eight hours production represented by the sample shall be rejected. The test shall be performed as specified in 4.3.1.

4.2.3.2 *Hydrostatic test (see 3.3)—Critical defect—Code No. 01006.* This test shall be conducted 100 percent in accordance with the procedure specified in 4.3.2. If any burster tubing fails to comply with the requirement specified, it shall be classed defective and no burster tubes shall be fabricated therefrom.

4.2.4 *Inspection Equipment.* Index of Inspection Equipment Number IEL–9205343 identifies the inspection equipment required to perform the examinations and tests prescribed in this section. The contractor shall design inspection equipment in accordance with the instructions in paragraph 6.3.

4.2.4.1 *Government rights to documentation.* Inspection equipment drawings and lists provided and revised in accordance with the requirements of the IEL may be used by DOD activities for design, procurement, manufacture, testing, evaluation, production and receiving inspection, overhaul, shipping, storage, identification of stock, ordering and storage of replacement parts, inspection of items at overhaul, general maintenance of equipment, construction, survey and whenever inspection equipment drawings are needed.

4.2.4.2 *Supply and maintenance.* Supply and maintenance of the equipment listed on the IEL shall be in accordance with MIL–I–45607.

4.2.4.3 *Government use of contractor's inspection and test equipment.* The contractor shall make available to the Government all inspection and test equipment necessary for

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determining conformance with contract requirements. Personnel for operating the equipment, and verification of its accuracy, shall be supplied by the contractor for the performance of examination or test by the Government.

4.3 Test methods and procedures.

4.3.1 *Determination of binder lubricant.*

4.3.1.1 *Stearic acid.* The sample shall be transferred to a tared medium porosity sintered-glass crucible. The sample shall be extracted with three 20 milliliter (ml.) portions of chloroform which has been previously saturated with RDX. The chloroform shall remain in contact with the residue for approximately 5 minutes before applying suction. The residue shall be stirred occasionally to break up any lumps which may persist and the sides of the crucible shall be washed down with each 20 ml. portion of chloroform to insure complete removal of the stearic acid. The crucible and contents shall be aspirated until the odor of chloroform is no longer detectable. The crucible and contents shall be dried in an oven maintained at 100 to 105 degrees Centrigrade (degrees C.) for one half hour, cooled in a desiccator and weighed. The above extraction, drying and weighing procedure with individual 20 ml. portions of chloroform shall be repeated until the loss in weight on subsequent extraction totals less than 0.003 grams. The percentage of stearic acid shall be calculated as follows:

Percentage of stearic acid = 100 A divided by W

where:

A = loss in weight of the crucible plus contents.

W = weight of sample.

4.3.1.2 *Graphite.* The residue remaining after the stearic acid has been removed shall be extracted with two 20 ml. portions of dimethyl formamide. The dimethyl formamide shall be maintained in contact with the residue for approximately 5 minutes with intermittent stirring before applying suction. The crucible and contents shall be washed with a min. of three 20 ml. portion of acetone. The acetone shall remain in contact with the residue for approximately 1 minute with continuous stirring before applying suction. The following qualitative test for RDX in the acetone washing shall be performed to determine complete extraction of the RDX. The acetone filtrate shall be transferred to a 20 ml. beaker and evaporated to dryness on a steam bath and then add several drops of a solution of chromotropic acid dissolved in 36 normal sulfuric acid. The formation of a reddish-brown color, as compared to a pink color obtained by a blank prepared in an identical manner, indicates the presence of RDX. The residue shall be aspirated until the odor of acetone is no longer detectable, then placed in an oven maintained at 100 degrees C. to 105 degrees C. for one half hour, cooled in a desiccator and weighed. The percent of graphite shall be calculated as follows:

Percentage graphite = 100 A divided by W

where:

A = increase in weight of crucible

W = weight of sample.

4.3.2 *Hydrostatic test.* Prior to machining, the burster tubing shall be tested for not less than five seconds at the minimum hydrostatic pressure as calculated by the following formula:

P = (2 St times .9) divided by D

where:

P = test pressure in pounds per square inch (psi)

S = minimum specified yield strength, psi

t = specified minimum wall thickness in inches

D = maximum outside diameter in inches

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5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.

5.1.1 *Level C.* The M19 projectile burster shall be preserved and packaged in accordance with dwg. 8799212.

5.2 Packing.

5.2.1 *Level C.* The cartons shall be packed in accordance with dwg. 8799213.

5.3 Marking. Marking shall be as specified on applicable packing dwg. and on dwg. 8796522.

5.4 Data cards. Data cards shall be prepared for each lot in accordance with the information specified in MIL–A–2550.

6. NOTES

6.1 Ordering data. Procurement documents shall specify title, number and date of this specification.

6.2 Inspection code numbers. The five digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government.

6.3 Inspection equipment. The contractor shall design inspection equipment as required by the referenced Inspection Equipment Lists (IEL) in accordance with the instructions of paragraphs 6.3.1 through 6.3.7.

6.3.1 *Inspection equipment lists.* Inspection equipment lists indicate the availability of inspection equipment designs by showing in the number column of the list of inspection equipment (00 Form 1242–3) the numbers of drawings of existing equipment designs or codes as indicated in paragraph 6.3.2. Design action required of the contractor with respect to the different types of drawings that may be listed is described in paragraphs 6.3.3 and 6.3.4. Action required by the contractor with respect to commercial inspection equipment is described in 6.3.5. The contractor will be required to prepare detailed drawings in accordance with 6.3.6 for all the equipment coded as "Contractor Design" in the number column. These contractor designs must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.7.

6.3.2 *Inspection equipment list codes.* The inspection equipment as defined in 6.3.3, 6.3.4, 6,3,5 and 6.3.6 will be designated in the IEL by the following codes:

CDAF — Contractor's design responsibility on Army format in accordance with MIL–D–45608.

CDCF — Contractor's design responsibility on contractor format.

AD — Army design.

ADMU — Army design, mandatory for use.

CE — Commercial equipment.

SCD — Specification control drawing.

6.3.3 *Army designs.* Army designs are reflected on detailed drawings which completely depict all the information necessary for the fabrication of the item of inspection equipment. The contractor need provide no design when an Army design is listed for an item of inspection equipment. Army designs fall into two basic classifications, mandatory and non-mandatory. When an inspection equipment list references mandatory Army designs, the contractor shall comply with, and use these designs accordingly. The contractor may, however, in connection with non-mandatory designs and with the approval of the Government, design alternate inspection equipment or use comparable commercial equipment to facilitate his operations. Such contractor prepared designs or commercial equipment selections must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.7.

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6.3.4 *Specification control drawing.* Specification control drawings depict the minimum equipment requirements in outline, descriptive, diagrammatic or pictorial form only and specify the required performance or other characteristics. Contractors must prepare detailed drawings (see 6.3.6) of their designs in support of specification control drawings. These contractor prepared designs must be approved by the Government prior to the fabrication or procuring of the equipment. Commercial equipment meeting the requirements of specification control drawings may be approved if described in sufficient detail to permit identification and evaluation by the Government. Designs shall be submitted for approval as specified in 6.3.7.

6.3.5 *Commercial equipment.* Commercial equipment is inspection equipment that has universal application for a specific function. It is comprised of items commonly used by industry and Government. Contractors are not required to furnish drawings of commercial inspection equipment, but a list of such equipment must be approved by the Government. Lists shall be submitted for approval to the inspection element of the agency administering the contract.

6.3.6 *Contractor designs.* Contractor designs are designs of inspection equipment for which the Government has assigned design responsibility to the contractor. Contractor designs shall be supported by detailed drawings which depict all information necessary to completely fabricate, calibrate and operate an item of inspection equipment. This requires that the necessary views, dimensions, materials, finish, notes, operating and calibration instructions be properly depicted in accordance with approved practices to the extent that further calculation or clarification will not be required. Contractor designs identified as CDCF may be developed on the format the contractor normally employs in his equipment design procedure provided such format reflects the detail and information specified above. Contractor designs identified as CDAF shall comply with the format and requirements of MIL–D–45608, and in addition, contain the detail and information specified above. Final approved contractor designs identified as CDCF shall be suitable for microfilming. They must be non-reproducible diazo or electrostatic prints of top quality full size with black line on white background and shall provide a legible print produced via the 35MM microfilm process. Contractor shall submit one copy of the final design as a rolled set or flat set.

6.3.7 *Submission of contractor designs.* Designs shall be submitted for approval to the Commanding Officer, Picatinny Arsenal, ATTN: SMUPA–ND. Design review will normally be accomplished within one month after receipt by Picatinny Arsenal. Partial submission of inspection equipment designs is permissible and encouraged. However, the Arsenal completion date for design review will be based on the date of the final submission of designs.

6.4 Supersession data. This specification supersedes in part MIL–C–10300, Charger, Burster, for Shell Loading, Assembling and Packing, dated 10 June 1952.

6.5 Hydrostatic test. If the burster tubing procured under this specification has been previously subjected to a hydrostatic test equivalent to that specified in this specification, and has successfully met the requirements, a second hydrostatic test will not be required.

Custodian:

Army—MU

Preparing activity:

Army—MU

Project No. 1310–A012

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